

IS1901

Microcontroller based ICT Project

Final report



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KICKBOTS ARENA

PRESENTED BY

OHMIES (GROUP - 12)

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245057J	Liyanage B.L.N.N
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Contents

1. Introduction	
2. Aim and Objective	
2.1 Aim	
2.2 Objective	
3. Problem and proposed solution	
3.1 Problem	
3.2 Solution and features	
4. Implementation	
4.1 Block diagram	
4.2 Schematic Diagram	
4.2.1 Arena	
4.2.2 Car	
4.2.3 Controller	
4.3 PCB Design	
4.4 3D view of the output	
4.3 Circuit Diagram	
4.5 Project procedure	
5. Components and used technologies	
5.1 Joystick controller	
5.2 Kick bot	
5.3 Gaming arena	
5.4 Software components	
6. Work Distribution	
6.1 245031 P.D.V.D.Gunasekara	
6.2 245056F Lathursan K	

6.3 245124M S.M.M.A.Madugalla

6.4 245057J - Liyanage B.L.N.N.

6.5 245060L-S.M.K.Madhushani

7. Issues Faced.....

8. Further Improvements.....

9.Budget.....

10. References.....

1. Introduction

KickBots Arena is an innovative 2-player robotic football system that blends embedded control, robotics, wireless communication, and computer vision into an interactive game experience.

Each player controls a striker robot for attacking (drives the ball toward the goal) and a goalkeeper robot for defense (moves along a rail to block goals)

A Web application acts as the main control and display system.

2.Aim and Objective

2.1 Aim

To design and develop a 2-player robotic football arena where each player controls a striker and goalkeeper robot using wireless communication and an interactive web interface, enabling a real-time, competitive and automated gaming experience.

2.2 Objectives

- Build striker and goalkeeper robots with precise movement
- Implement wireless control using joysticks and nRF24L01 modules
- Create a web-based interface to display timer, match status, and live score updates
- Detect goals automatically using IR sensor modules
- Control goalkeeper movement on a linear rail system using NEMA 17 motors
- Ensure all microcontrollers (Arduino, ESP32) communicate reliably in real-time
- Develop a reset and match-restart functionality with scoring history
- Enable a smooth multiplayer experience with minimal delay

3. Problem and proposed solution

3.1 Problem

Traditional tabletop and arcade football games offer limited interaction, fixed player motion, and lack automation. Players cannot control multiple actions such as shooting or goalkeeping realistically. Scoring is also manual and subjective, which can lead to errors and disputes. In addition, most hobby robotic games are single-player and do not provide real-time competition between two players. There is a lack of an engaging, automated robotic football system that simulates real gameplay with accurate scoring and digital control.

3.2 Solution and features

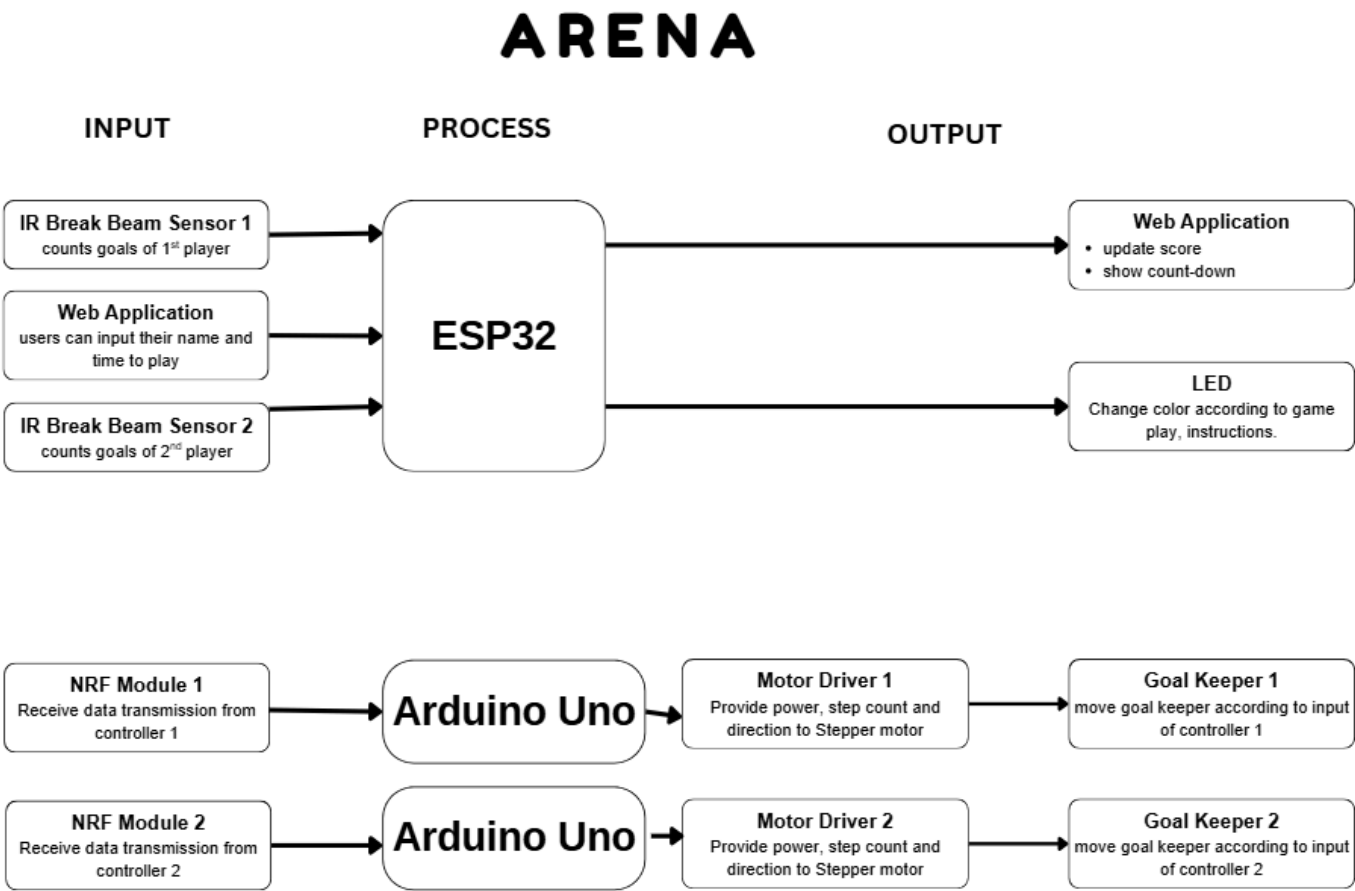
KickBots Arena introduces a fully automated 2-player robotic football system controlled through wireless joysticks and displayed on a web interface.

1. Each player controls two robots - a striker with a kicking mechanism and a goalkeeper on a motorized rail.
2. IR sensors detect goals automatically, and scores update in real-time on a web app.
3. The web interface manages match time, start/pause/reset, and announces the winner.
4. Wireless modules ensure smooth and responsive gameplay.

This solution transforms robotic football into an automated, interactive, competitive experience with accurate score tracking and multi-device control.

4. Implementation

4.1 Block diagram

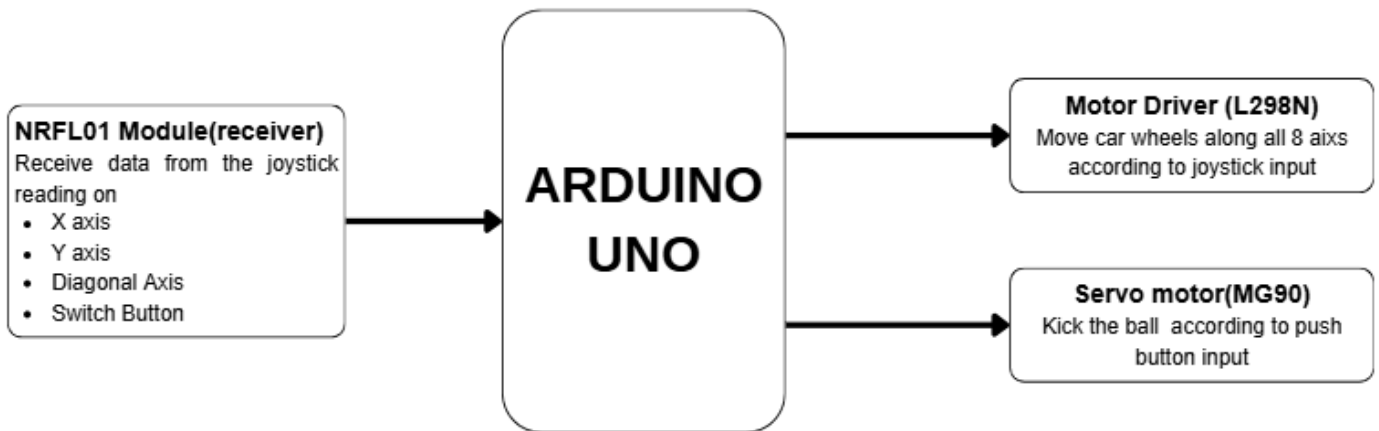


CAR

INPUT

PROCESS

OUTPUT

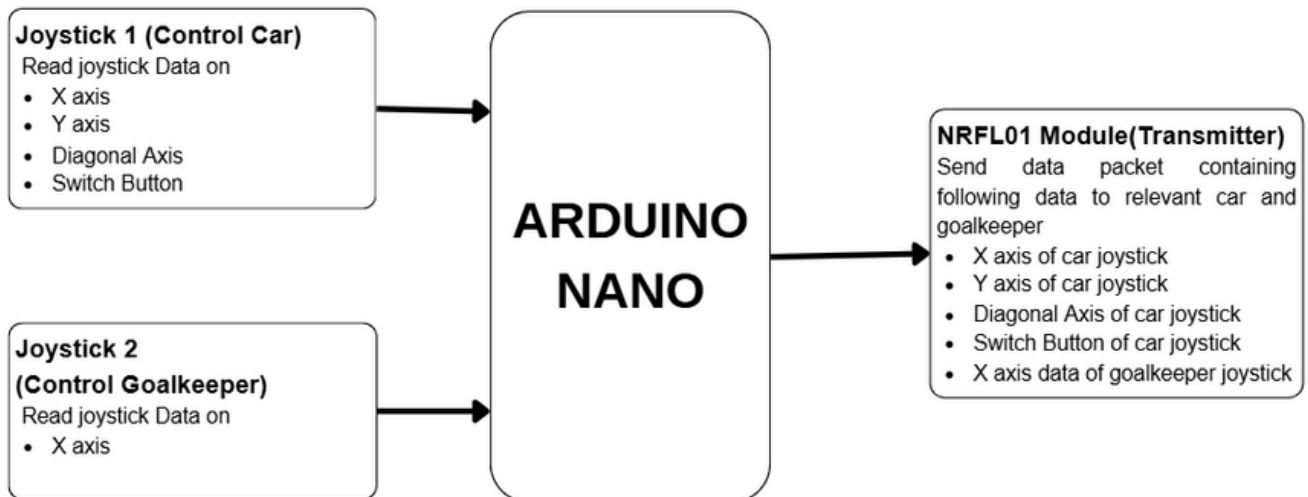


CONTROLLER

INPUT

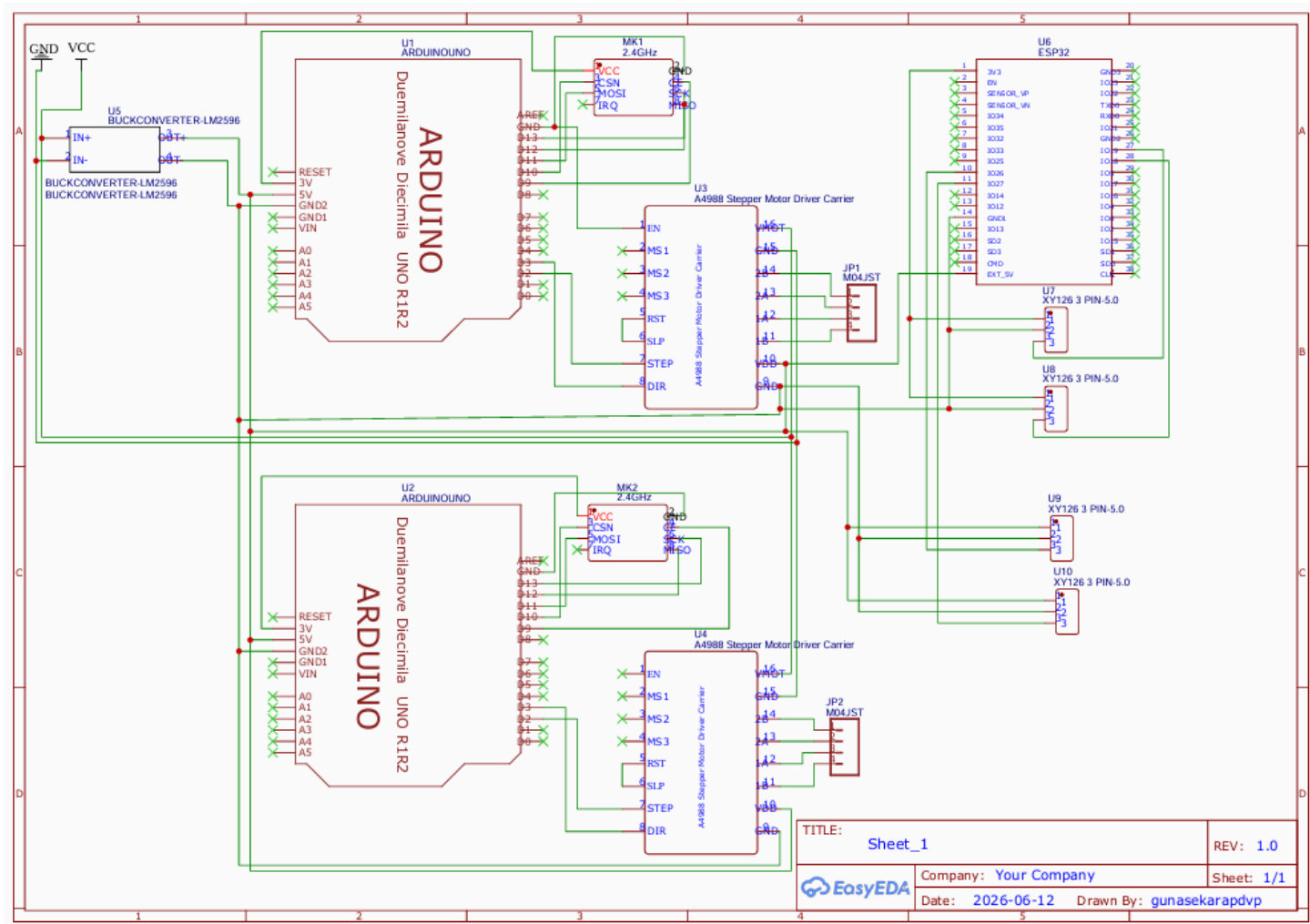
PROCESS

OUTPUT

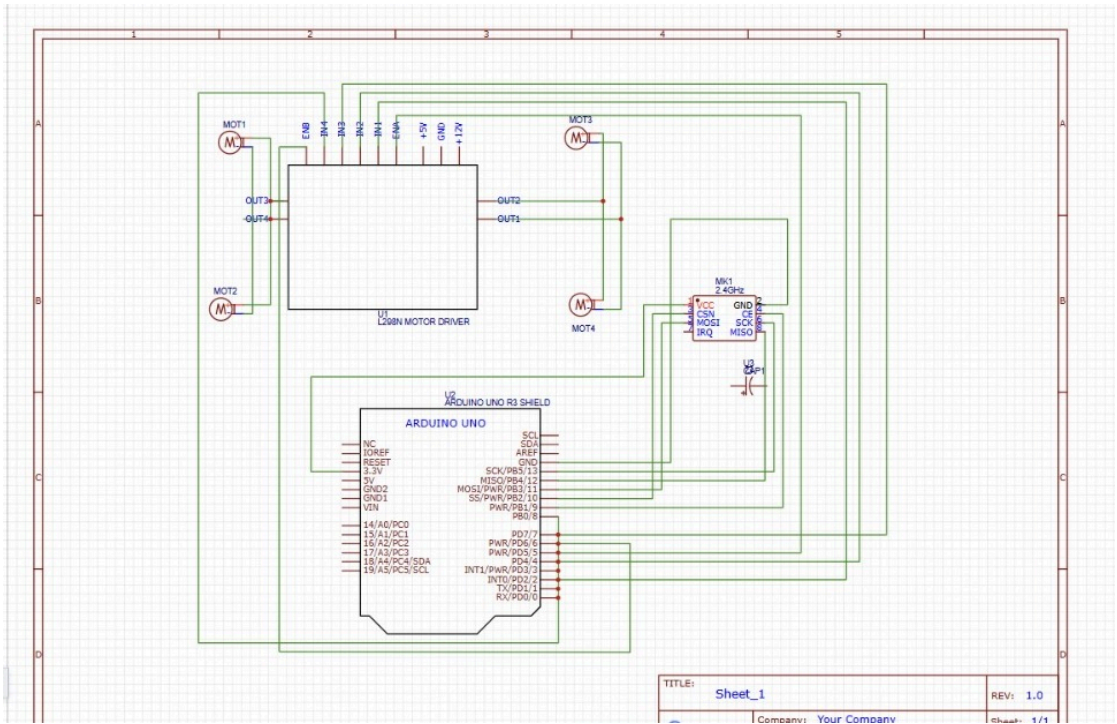


4.2 Schematic Diagrams

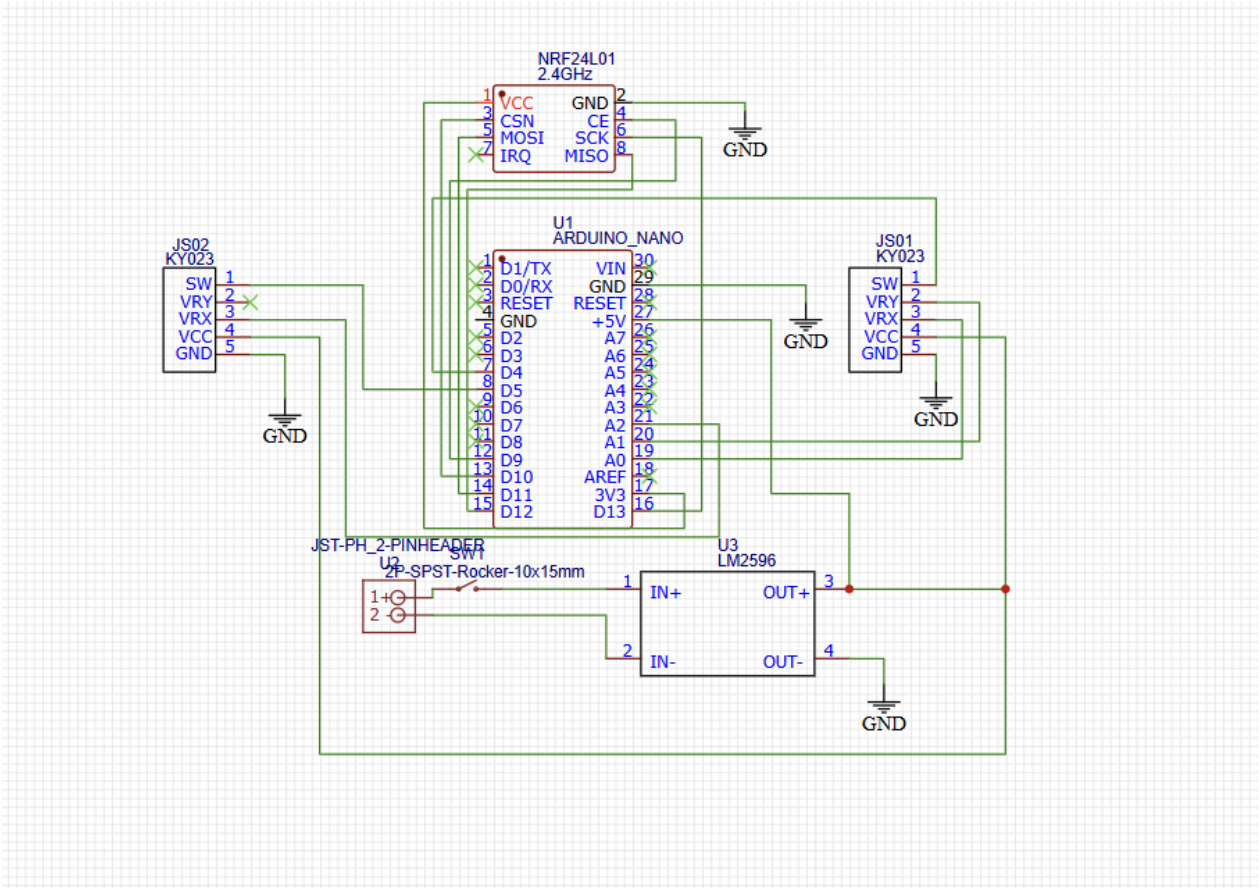
4.2.1 Arena

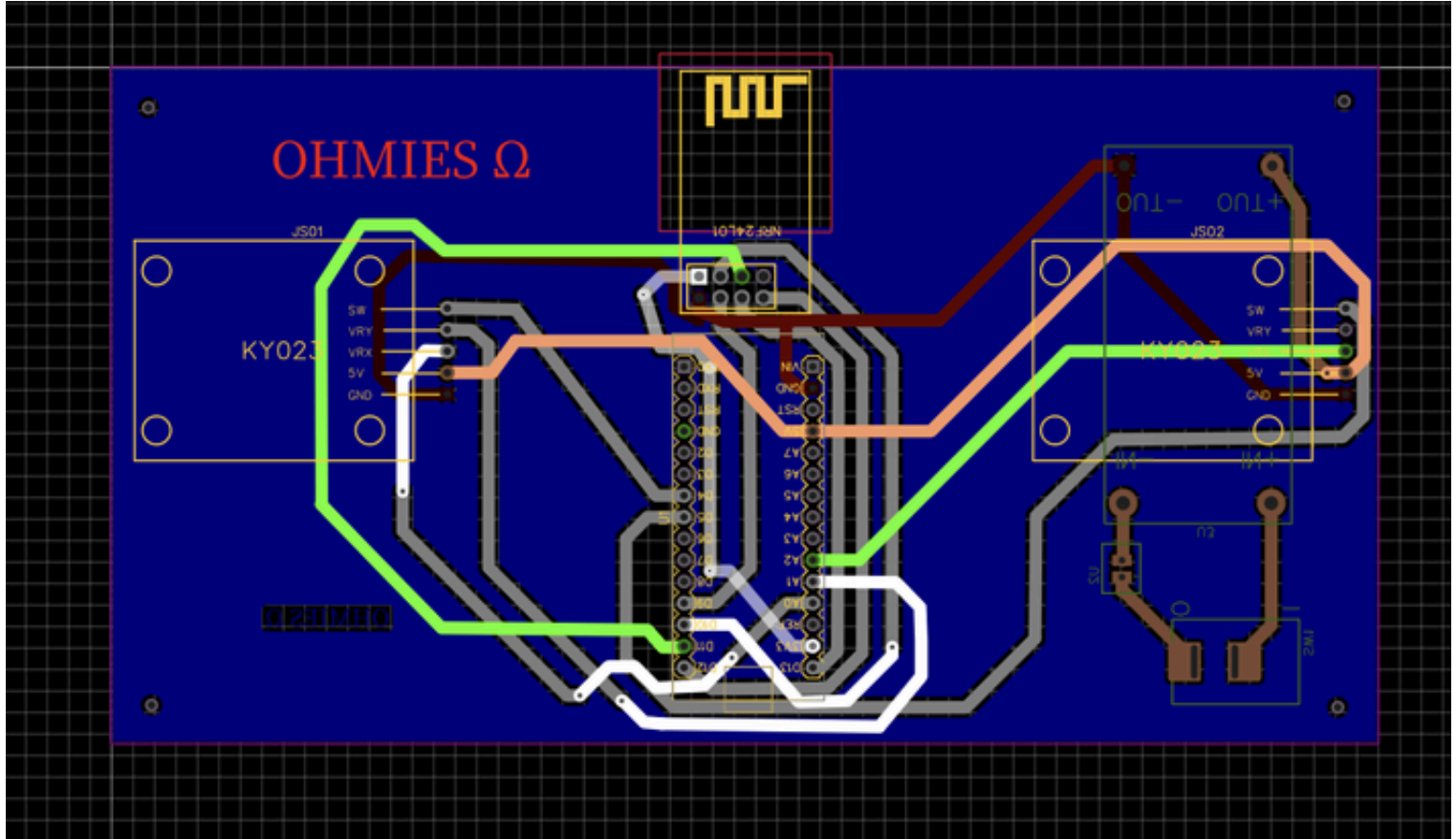


4.2.2.Car

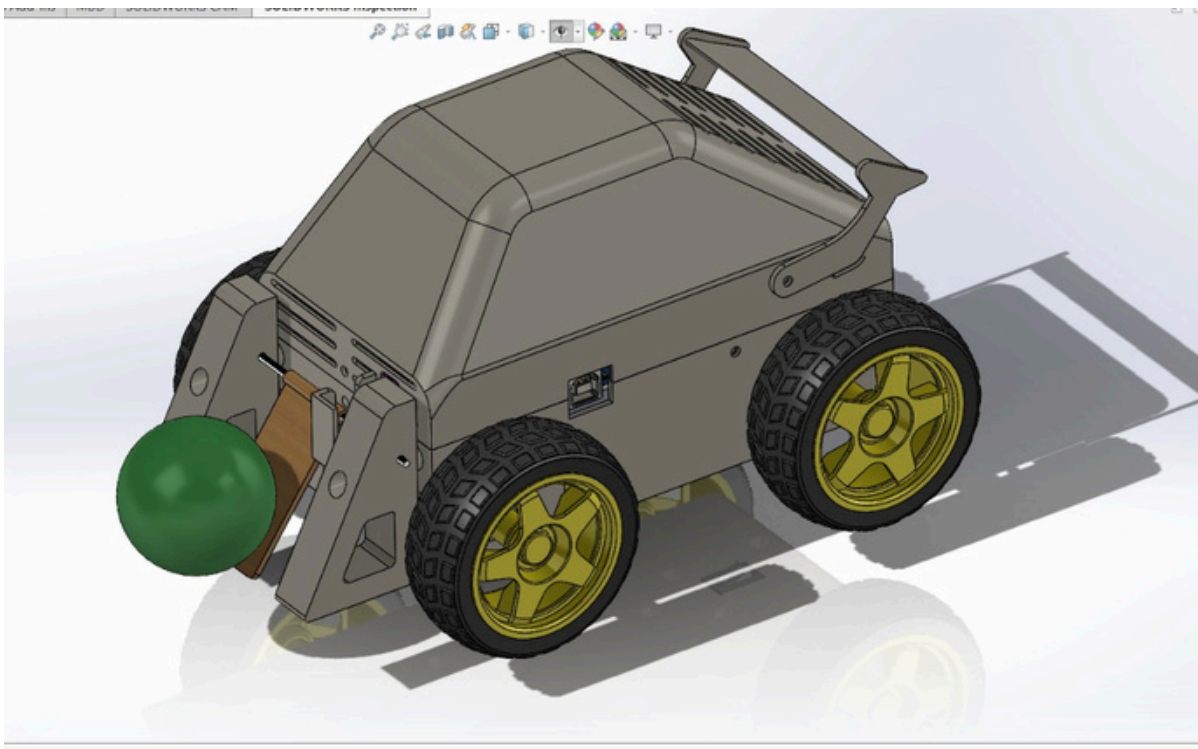
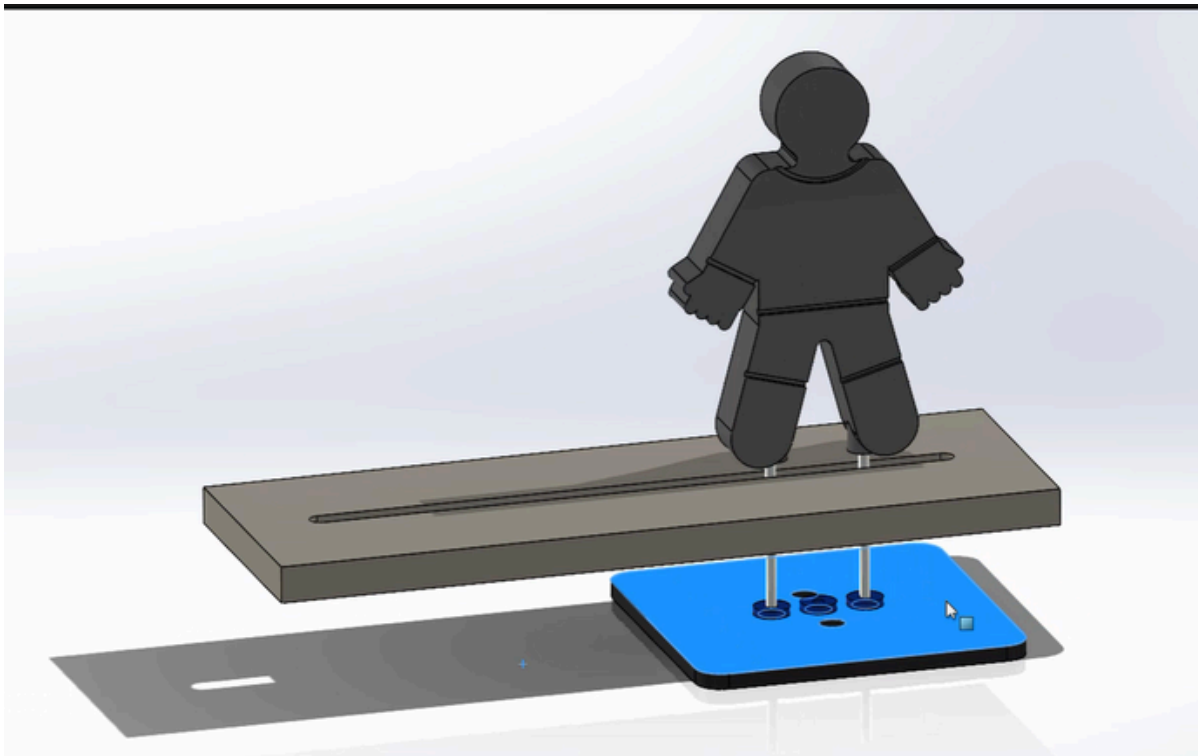


4.2.3. Controller

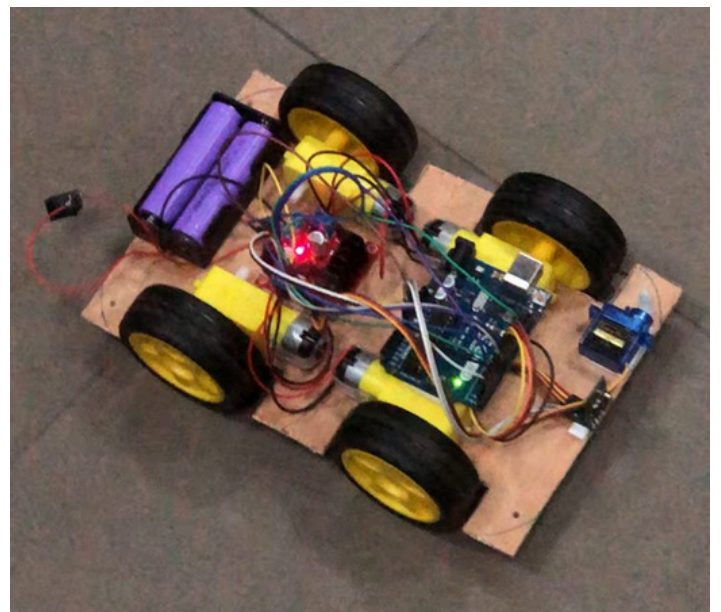
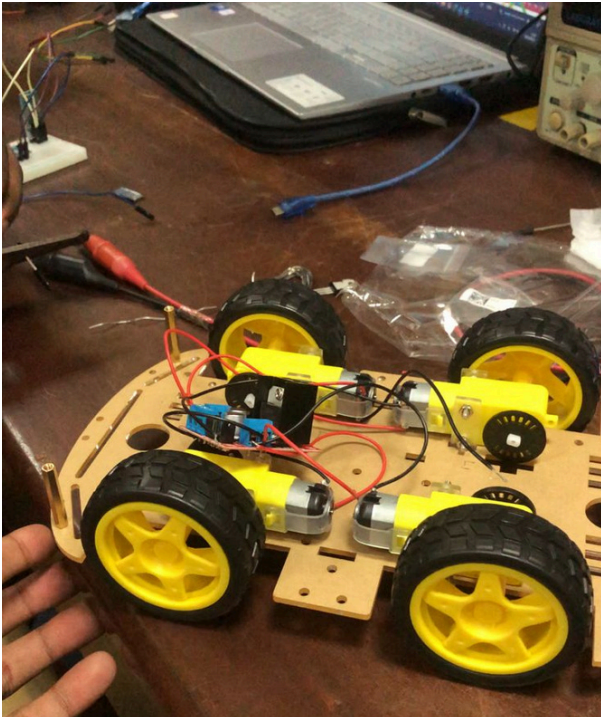


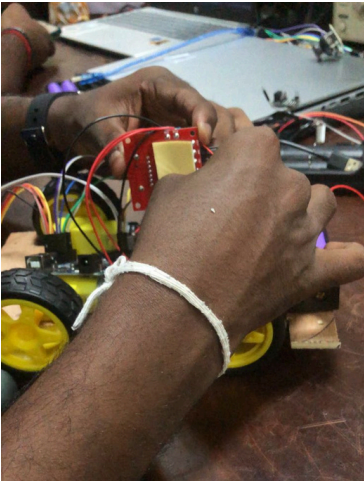
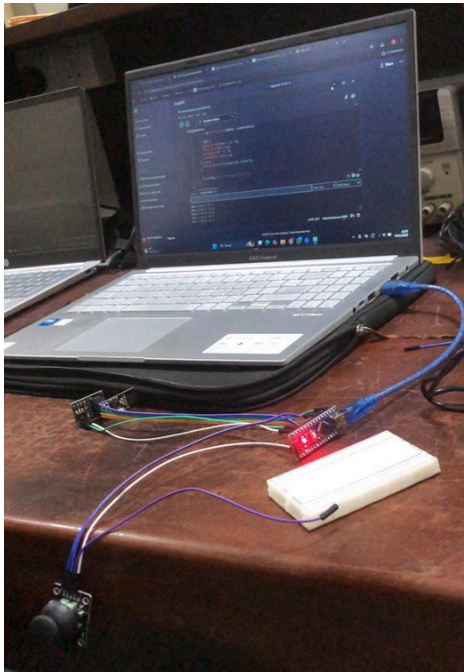


4.4 3D Design



4.5 Project procedure





5. Components and used technologies

5.1 controller

Joysticks



The two joysticks are used to control the robot.

The left joystick moves the car forward, backward, left, and right.

The right joystick can be used for extra controls, like moving a servo or showing input on the arena.

Arduino Nano



The Arduino reads the joystick positions (as analog values) and converts them into digital data.

It then sends this data to the nRF24L01 module using SPI communication.

nRF24L01 Transmitter



This module sends the joystick data wirelessly to the receiver on the car and another on the arena.

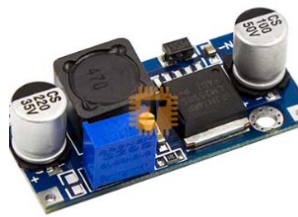
This allows both to respond in real time without any wired connection.

Battery_



The Li-ion battery powers the Arduino Nano and all connected parts, making the controller portable and easy to carry.

LM2596S Buck Converter



When the joysticks are moved, the Arduino reads the movement, sends the values wirelessly, and the car or arena reacts accordingly — for example, the car moves, or the arena display updates

Overall Function

When the joysticks are moved, the Arduino reads the movement, sends the values wirelessly, and the car or arena reacts accordingly — for example, the car moves, or the arena display updates

5.2 Kick bot

nRF24L01 Receiver



This module receives the wireless data sent from the controller.

The received data contains joystick positions that decide the car's movement.

Arduino Uno



The Arduino Uno on the car reads the data from the nRF24L01 module and decides which direction the motors should move.

It acts like the brain of the car.

Motor Driver (L298N).



The motor driver receives control signals from the Arduino and powers the DC motors accordingly.

It allows the car to move forward, backward, left, right, and diagonal axes depending on joystick input.

BO Motors



These motors are connected to the wheels and perform the actual movement of the car.

When the joystick is pushed forward, both motors move forward; when pushed back, both move backward; and for turns, only one motor runs.

Battery Pack

The Li-ion battery supplies power to the Arduino, motor driver, and motors.

It provides enough current for smooth movement.

Overall Function

When you move the joystick on the controller, the car instantly moves in that direction.

For example:

Joystick forward → Car moves forward

Joystick back → Car moves backward

Joystick left/right → Car turns left/right

Joystick Diagonal → Car moves along the Diagonal Path

5.3 Gaming arena

ESP32



The ESP32 controls the stepper motor and communicates wirelessly with the controller or through the web app.

It decides how and when the goalkeeper should move.

nRF24L01 Receiver



This module receives data from the controller, just like the car does.

The ESP32 reads this data to know the joystick movement.

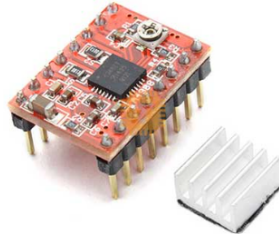
Stepper Motor and Timing Belt



The stepper motor moves the goalkeeper left and right smoothly along the rail.

The motor turns in small steps, allowing accurate movement.

A4988 Stepper Motor Driver



Web Application

The ESP32 connects to Wi-Fi and shows a web page that can display movement or allow manual control from a phone or laptop.

Power Supply



Power Adaptor gives power to the ESP32 and the stepper motor.

5.4 Software components

1. Arduino IDE

Used to write, upload, and debug the code for the ESP32 and Arduino Nano boards.

The programming is done in C/C++ language using Arduino libraries.

It includes the code for:

Reading joystick and sensor data

Sending and receiving data through the nRF24L01 module

Controlling the stepper motor for the goalkeeper

Handling serial communication

2. ESP32 Web Server

The ESP32 acts as a local web server that hosts the web application.

The web page is created using HTML, CSS, and JavaScript, stored in the ESP32's memory.

When a phone or laptop connects to the ESP32's Wi-Fi, it can open the web page

3. HTML / CSS / JavaScript (Web App)

These are used to design and run the web interface:

HTML – Structure of the web page (buttons, text, sliders)

CSS – Styling and layout for a clean design

JavaScript – Adds interaction (sending commands to ESP32 using HTTP requests)

The interface is simple and user-friendly so anyone can control the arena easily.

4. Libraries Used

WiFi.h – For connecting ESP32 to Wi-Fi.

WebServer.h or ESPAsyncWebServer.h – For hosting the web app.

AccelStepper.h – For controlling the stepper motor movement smoothly.

SPI.h and RF24.h – For communication between ESP32 and nRF24L01 module.

Arduino.h – For general Arduino functions and timing.

5. Overall Software Function

The controller sends joystick data wirelessly.

The ESP32 receives that data through Wi-Fi.

Based on the nRF24L01 data, the stepper motor moves the goalkeeper.

The web application displays goal scored and countdown.

6. Work Distribution

Team member	Role	Responsibility
245031B	Cars design and Game Logic	Arduino Uno Setup for Both cars, Design Chassis, Data Communication through NRF24L01 modules, Initiate Car movements along all directions, implementing the kickers for the cars. Implementing game logics
245056F	Controllers Design and Coding	Arduino Nano–based controllers for both players, implementation of NRF24L01 communication between the controllers, cars, and arena. IR sensor–based goal detection and counting system, ESP32 for web application and arena updates, system testing and debugging.
245060L	Other arena functions and Coding	Power supply for arena , car and controllers, Implementing other arena outputs (LED, Speaker Module)that change according to game play. App Development: Building App structure and UI.
245057J	PCB design and Goal counting mechanism	Controller PCB Design. Implementing goal counting mechanism for both ends .Connecting ESP32 of arena with web application(ESPNOW). Overall system testing, debugging and error correction.
245124M	Structure and Goal Keeper mechanism	Creating and design the arena to house all components along with both goal keeper mechanism, design Goal keeper movement (by NEMA 17 motors and rails/belts). Implementing limit switches for goal keepers.

6. Work Distribution

6.1 245031 P.D.V.D.Gunasekara

1. Arduino Uno Setup for Both Robot Cars

Configured and programmed the Arduino Uno microcontrollers used in both football robots. This included assigning pin connections, integrating motor drivers, kicker mechanisms, NRF24L01 communication modules, and ensuring proper interaction between all hardware components.

2. Robot Chassis Design and Assembly

Designed and assembled the mechanical structure of the football robots. The chassis was optimized to provide stability, durability, and sufficient space for electronic components while maintaining maneuverability during gameplay.

3. Supported in developing Wireless Communication Using NRF24L01 Modules

Supported team member in charge of Implementing the wireless communication system between controllers and robot cars using NRF24L01 transceivers. Developed and tested reliable data transmission protocols to ensure low-latency control during matches.

4. Development of Omnidirectional Car Movement

Programmed and tested robot movement algorithms to enable forward, backward, left, right, and rotational movements. Fine-tuned motor control parameters to improve responsiveness and driving accuracy.

5. Implementation of Core Game Logic

Developed core functionalities required for gameplay, including control processing, and supported in command interpretation, system coordination, and interaction between different subsystems of the project.

6. Coordination of Goalkeeper Movement Mechanism

Collaborated with team member responsible for the goalkeeper system by assisting in motion control planning, integration strategies, and testing procedures to ensure smooth goalkeeper operation.

7. Team Leadership and Project Coordination

Served as the group leader by coordinating project activities, monitoring progress, facilitating communication among team members, organizing testing sessions, and assisting in problem-solving throughout the development process.

6.2 245056F Lathursan K

1. Controller Design and Coding

Designed and coded Arduino Nano–based controllers for both players. Integrated joysticks, buttons, and other control inputs. Implemented dual joystick input for car's 8-direction movement and goalkeeper control.

2. Wireless Communication Using NRF24L01 Modules

Designed and implemented a complex multi-node wireless network using 6 NRF24L01 modules (4 receivers, 2 transmitters) for communication between controllers, RC cars, and the arena. To manage the complexity of multiple devices transmitting simultaneously, configured unique addresses for each module to distinguish individual components and prevent data collisions. Conducted extensive range and reliability testing to achieve stable performance throughout the arena.

3. Power Management

Worked on power distribution for both wireless controllers using 2 Li-ion batteries with buck converters to regulate stable 5V output for Arduino Nano and dual joysticks. Ensured clean and consistent power delivery for reliable controller operation during extended gameplay sessions.

Assisted in designing power management for the arena using 12V 2A supply for both Arduino Uno boards and ESP32. Implemented 12V 5A supply to power remaining arena components including LED indicators, NEMA 17 stepper motor for goalkeeper, and IR break beam sensors. Ensured adequate current distribution across all subsystems without voltage drops.

4. Assisted Web Application Coding and Integration

Assisted in connecting the ESP32 with the web application for real-time updates. Contributed to building the live scoreboard and match status display. Supported verification of communication between the LED, IR sensors and web interface.

5. Supported Core Game Logic and Goalkeeper

Collaborated with team member in developing core functionalities required for gameplay including control processing, command interpretation, and system coordination. Assisted in coding and testing the goalkeeper movement mechanism and supported integration strategies for smooth operation. Ensuring the functionality and responsiveness of the goal keeper with controller through NRF24L01.

6. Supported assembling Arduino Uno Setup for Both Robot Cars

Assisted the member with assembling the car. Also in checking the functionality and responsiveness of the car with controller through NRF24L01. This included assigning pin connections, integrating motor drivers, kicker mechanisms.

7. System Testing and Debugging

Integrated controller, arena, and ESP32 functionalities. Performed testing and troubleshooting of hardware and software components. Identified and resolved motor direction faults, radio failsafe gaps, and power instability to ensure reliable gameplay.

6.3 245124M S.M.M.A.Madugalla

1. Goalkeeper Mechanism Development

Designed and assembled a two-rail linear slide system with a NEMA 17 stepper motor and GT2 timing belt to enable precise horizontal goalkeeper movement.

2. Goal Detection System

Implemented an IR break-beam sensor-based goal-counting mechanism and configured ESP32 interrupts for real-time goal detection.

3. Audio Feedback Integration

Installed and integrated speakers to provide sound effects and commentary cues during gameplay events.

4. Arena Assembly and Integration

Assisted in assembling the arena structure, goal areas, and power distribution network for reliable system operation.

5. Robot Chassis Design and Assembly Support

Contributed to the design, construction, and integration of the robot chassis and its mechanical components.

6.4 245057J - Liyanage B.L.N.N.

1. PCB Design

Designed the complete controller circuitry for the wireless control unit by integrating dual-axis joystick modules, an Arduino Nano microcontroller, an NRF24L01 wireless communication module, and the required power management circuitry into a unified schematic. Ensured proper signal flow, component compatibility, and stable power distribution to support reliable controller operation.

2. PCB Layout and Fabrication

Developed the PCB layout for the controller, optimizing component placement and signal routing to achieve a compact and efficient design. Incorporated the necessary power regulation circuitry and communication interfaces while minimizing potential signal interference. Participated in PCB assembly, testing, and validation to ensure reliable joystick input processing and wireless communication performance.

3. Development of Goal Detection and Web Integration Module

Developed the goal detection subsystem by integrating IR break-beam sensors with the ESP32 microcontroller and the web application. Implemented real-time goal detection, data processing, and communication between the ESP32 and the software platform to enable automatic score updates and goal notifications. Conducted testing and debugging to improve detection accuracy and ensure reliable synchronization between hardware and software components.

4. Collaboration with Car Integration

Collaborated with the team member responsible for Car 2 during the system integration phase. Assisted in validating wireless communication between the remote controller and the Arduino Uno on Car 2 to ensure responsive and low-latency movement execution. Supported signal mapping and control parameter adjustments to ensure joystick inputs were accurately translated into motor speed and direction commands.

5. Collaboration with Goalkeeper System Integration

Collaborated with the goalkeeper subsystem team during integration and testing activities. Assisted in analyzing joystick input behavior and its relationship with the goalkeeper movement mechanism. Supported the testing and refinement of movement synchronization to ensure smooth and accurate horizontal positioning along the goalkeeper rail system.

6. Collaboration with Arena Development and Fabrication

Contributed to the fabrication and preparation of arena components used in the final system. Assisted in assembling, refining, and polishing various game elements to improve mechanical interaction between the robots, playing field, and goals. Participated in testing sessions to ensure smooth gameplay operation and overall system reliability.

6.5 245060L-S.M.K.Madhushani

1.Power Supply Distribution Network (PDN) Design

- **Dual-Bus Isolation Strategy:** Engineered to completely decouple high-current inductive and resistive loads from sensitive digital logic circuits.
- **Arena Dual-SMPS Setup:** * **12V 10A SMPS:** Dedicated entirely to the WS2812B LED strips to handle high current draw during peak brightness without causing voltage drops on the processing line.
 - **12V 2A SMPS:** Acts as an isolated logic power supply for the central ESP32 and Arduino Uno microcontrollers.
- **Goalkeeper Defense Rail Powering:** Routes continuous 12V power directly to the NEMA 17 stepper motor, while utilizing an onboard step-down DC-DC Buck Converter to supply a regulated 5V to the logic pins of the A4988 stepper driver.
- **Striker Car Power Source:** Powered completely independently using high-capacity, mobile 18650 Lithium-Ion cells.
- **Handheld Remote Controllers:** Uses a single 18650 battery cell stepped up to a continuous 5V via a DC-DC boost converter to run the Arduino Nano and dual-axis analog joysticks.
- **Wireless Transceiver Safeguard:** Feeds a precise, ripple-free $\$3.3\text{V}$ supply directly to the power-sensitive nRF24L01 wireless module to protect against hardware damage and maintain telemetry signal integrity.

2.Interactive Arena Gameplay Outputs

- **Real-Time Feedback Systems:** Synchronizes visual and acoustic stadium status indicators directly with the active backend game loop.
- **WS2812B Dynamic LED Strips:** Arrays waterproof addressable RGB LED strips around the stadium walls operated by a finite state machine firmware layout.
- **Visual Match State Indication:** Automatically illuminates the pitch walls in continuous green upon game initiation, switching instantly to a flashing red pattern the exact millisecond an IR break-beam sensor registers a goal.
- **Acoustic Signaling Loop:** Outputs localized stadium sound effects (Match Start, Goal Detection, and Match End) directly through the host computer's built-in speaker system.

3.Web App Architecture & UI/UX Development

- **Static IP Wi-Fi Communication:** Establishes wireless web application communication over a local Wi-Fi local area network (LAN) pointing directly to the unique IP address hosted by the arena's ESP32.
- **Asynchronous Telemetry Updates:** Routes sensor messages via the wireless gateway when a goal event breaks the infrared beam, allowing the web UI scoreboard to dynamically increment score counts and run down the match timer smoothly without requiring full webpage refreshes.

7.Issues Faced

During the development of the Football Robot Arena project, several technical and practical challenges were encountered. These issues required troubleshooting, testing, and design modifications to ensure the successful completion of the system.

1. NRF24L01 Communication Instability

- Establishing reliable wireless communication between the controllers and robot cars was initially difficult.
- Signal interruptions and inconsistent data transmission occurred due to power fluctuations and improper module connections.
- Additional testing and optimization of the communication setup were required.

2. Power Supply Management

- Different components required different operating voltages, making power distribution challenging.
- Voltage drops affected the performance of the NRF24L01 modules and motor drivers.
- Buck converters were incorporated to provide stable voltage levels.

3. Motor Control and Movement Accuracy

- Achieving smooth movement and accurate steering of the robots required multiple adjustments to motor control logic.
- Differences in motor speeds caused the robots to drift from their intended paths.
- Software calibration and hardware testing were performed to improve maneuverability.

4. Goal Detection Reliability

- The IR break beam sensors occasionally produced false detections due to ambient light interference and alignment issues.
- Proper positioning and calibration of the sensors were necessary to ensure accurate goal detection.

5. ESP32 Web Interface Development

- Configuring the ESP32 to create its own web server and communicate effectively with arena components required extensive testing.
- Network connectivity issues occasionally affected the accessibility of the control interface.

6. Goalkeeper Mechanism Synchronization

- Coordinating the movement of the goalkeeper mechanism with user inputs presented timing challenges.
- Adjustments to control logic and motor operation were necessary to achieve responsive movement.

7. Hardware Integration Challenges

- Integrating multiple modules, including motor drivers, wireless communication modules, sensors, LEDs, and controllers, increased wiring complexity.
- Troubleshooting connection errors and ensuring compatibility between components consumed significant development time.

8. Mechanical Design Constraints

- Space limitations within the robot chassis made component placement difficult.

- Several redesigns were required to accommodate batteries, motors, controllers, and communication modules while maintaining stability and performance.

Despite these challenges, systematic testing, debugging, and iterative improvements enabled the successful implementation of the Football Robot Arena system.

8. Further Improvements

Although the Football Robot Arena system successfully achieved its intended objectives, several enhancements can be implemented in future versions to improve performance, reliability, and user experience.

1. Improved Wireless Communication

- Upgrade the communication system to use more reliable wireless technologies with greater range and reduced interference.
- Implement error-checking mechanisms to ensure accurate data transmission between controllers and robots.

2. Enhanced Robot Mobility

- Utilize higher-quality motors and motor drivers to achieve smoother movement and better control.
- Incorporate speed control algorithms for more precise navigation and maneuverability.

3. Autonomous Gameplay Features

- Integrate additional sensors such as ultrasonic, vision, or camera modules to enable autonomous ball tracking and obstacle avoidance.
- Develop artificial intelligence algorithms that allow robots to make independent gameplay decisions.

4. Advanced Goal Detection System

- Improve goal detection accuracy by using multiple sensors or sensor fusion techniques.
- Implement automatic fault detection to identify sensor malfunctions.

5. Mobile Application Integration

- Develop a dedicated mobile application for monitoring scores, controlling game settings, and viewing match statistics.
- Enable wireless configuration of arena settings through the application.

6. Battery Management System

- Incorporate battery monitoring features to display real-time battery levels for both robots and controllers.
- Add low-battery alerts to prevent unexpected shutdowns during gameplay.

7. Enhanced Arena Features

- Introduce a digital scoreboard with match timers and player statistics.
- Add sound effects, animations, and customizable lighting patterns to improve user engagement.

8. Improved Mechanical Design

- Design a more compact and lightweight robot chassis for increased speed and agility.
- Utilize stronger and more durable materials to improve long-term reliability.

9. Automatic Goalkeeper System

- Develop an intelligent goalkeeper capable of automatically detecting and blocking incoming shots.
- Implement predictive movement algorithms to improve defensive performance.

10. Internet Connectivity and Data Analytics

- Connect the system to cloud platforms for storing match data and player statistics.
- Provide performance analysis and historical records through a web dashboard.

9.Budget

Component	Unit Price (Rs.)	Quantity	Total Price (Rs.)
Geared Motors	130	8	1040
Arduino uno	2000	4	8000
Nrf	350	6	2100
L298N motor driver	420	2	840
switch	50	4	200
car 3D	6000	2	120000
Arduino nano	1300	2	2600
Joystick Module	420	4	1680
LM2596 Buck converter	150	4	600

battery 3.7V	500	10	5000
PCB		2	9000
NEMA 17 stepper motor	2620	2	5240
A4988 steppper motor driver	280	2	520
ESP 32	1500	1	1500
Power supply Unit	1900	1	1900
Goal Keeper 3D	900	2	1800
Rail(Extrusion bar, gantry plate& etc)	7000	2	14000
Melamaine board & cutting			12000
Total Budget			80020

10. References

https://youtu.be/Z7_Cy66Vnrc?si=KXOrc6KhChikbLDL

<https://youtu.be/JSHJ-RLbNjk?si=hRrSfhF58SNRVxCc>

<https://forum.arduino.cc/t/nrf24l01-ping-pong-issue/559350>

<https://github.com/aslam-Hossain-YT/Joystick-Control-Bluetooth-Car>

<https://youtu.be/ENRFh83hHM8?si=avthUNHjc7eUXwmE>